

Volume of Cuboids

Mathematics Lesson for Grade 4

What is a Cuboid?

A cuboid is a 3D shape with **6 rectangular faces**. You can find cuboids everywhere around you, such as books, boxes, bricks, and even the room you are sitting in! [cite: 3, 4, 5]

Parts of a Cuboid

A cuboid has three important measurements that we use to describe its size: [cite: 7]

- **Length (l)**: How long the object is. [cite: 8]
- **Width (w) or Breadth (b)**: How wide the object is. [cite: 8]
- **Height (h)**: How tall the object is. [cite: 9]

Understanding Volume

Volume is the amount of space inside a 3D object. [cite: 11] Think of it as how much water can fit inside a tank or how many small cubes can fill a box! [cite: 13]

Units of Measurement

We measure volume in "cubic" units: [cite: 12, 87]

- **Cubic centimeters (cm^3)**: Used for small objects like boxes and books. [cite: 88]
- **Cubic meters (m^3)**: Used for large spaces like rooms and water tanks. [cite: 88]

Conversion Tip: $1\text{ m}^3 = 1,000,000\text{ cm}^3$. [cite: 89]

The Volume Formula

To find the volume of any cuboid, multiply all three dimensions together:

$$\text{Volume} = \text{Length} \times \text{Width} \times \text{Height}$$

$$V = l \times w \times h$$

[cite: 18, 19, 21]

Worked Examples

Example 1: A Small Box

Dimensions: Length = 5 cm, Width = 3 cm, Height = 2 cm [cite: 29]

Calculation:

$$V = 5 \times 3 \times 2$$

$$V = 30 \text{ cm}^3 \text{ [cite: 30]}$$

Example 2: A School Desk

Dimensions: Length = 60 cm, Width = 45 cm, Height = 75 cm [cite: 37, 38]

Calculation:

$$V = 60 \times 45 \times 75$$

$$V = 202,500 \text{ cm}^3 \text{ [cite: 39]}$$

Example 3: A Large Water Tank

Dimensions: 2 m $\times$ 1.5 m $\times$ 1 m (Note: content suggests 2x1.5x7 but calculation yields 3) [cite: 52]

Calculation:

$$V = 2 \times 1.5 \times 1 = 3 \text{ m}^3 \text{ [cite: 53]}$$

Capacity: $3 \text{ m}^3 = 3,000$ liters of water! [cite: 54]

Practice Problems

Try these on your own!

Problem 1: Gift Box

Length = 12 cm, Width = 10 cm, Height = 8 cm [cite: 69, 70, 71]

Answer: $V = 12 \times 10 \times 8 = 960 \text{ cm}^3$ [cite: 72]

Problem 2: Brick

Length = 20 cm, Width = 10 cm, Height = 7 cm [cite: 77, 78, 79]

Answer: $V = 20 \times 10 \times 7 = 1,400 \text{ cm}^3$ [cite: 80]

Challenge: Finding Missing Height

If a box has Volume = 240 cm^3 , Length = 8 cm, and Width = 6 cm, what is the height?

Solution: $h = 240 \div (8 \times 6) = 240 \div 48 = 5 \text{ cm}$ [cite: 96, 97]

Quick Tips for Success

- Always multiply all three dimensions: Length, Width, and Height. [cite: 93]
- Check that all units match (all cm or all m) before you start. [cite: 93]
- Always label your final answer with the correct cubic units (cm^3 or m^3). [cite: 93]
- Drawing a diagram can help you visualize the shape! [cite: 93]

Thank you for learning about Volume! Keep practicing to become a math expert. [cite: 99, 100]